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Group Project
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Predictors of Survival Outcomes Among Post-Transplantation Acute Leukemia Patients: An Exploratory Evaluation of Patient, Donor, and Clinical Factors

Introduction

A standard treatment for patients with acute leukemia is bone marrow transplantation, with the goal of long-term remission and cure. However, recovery following transplantation is a complex and dynamic process, and may be influenced by both baseline patient and donor characteristics and clinical events that occur during the post-transplantation period. Therefore, understanding which factors affect outcomes is necessary for improving patient prognosis and guiding clinical management.

The prognosis for a patient after transplantation may be influenced by baseline characteristics, such as patient and donor age and sex, disease stage, cytomegalovirus (CMV) immune status, leukemia disease classification, and prophylactic use of methotrexate to prevent acute graft-versus-host disease (aGVHD). Additionally, post-transplant events, such as the development of aGVHD and recovery of platelet counts to normal levels and the development of aGVHD may modify the risk of relapse or death.

The study used the baseline and longitudinal data (listed above) collected from a cohort of 137 enrolled patients, who underwent allogeneic bone marrow transplantation between March 1984 and June 1989, at four hospitals in the United States and Australia. All patients received a radiation-free conditioning regimen consisting of oral Busulfan and intravenous cyclophosphamide. Patients were followed up until death or the end of the study period.

The components and primary scientific questions addressed in this analysis were:

- Baseline and descriptive analyses:
    - What are the main characteristics of disease-free survival time?
    - How do patients across disease classifications compare, across baseline measurements?
    - Are baseline variables associated with differences in disease-free survival?
- aGVHD analyses:
    - Does aGVHD appear to be an important prognostic event?
    - Among the patients who develop aGVHD, are any of the baseline factors associated with differences in disease-free survival?
    - Is prophylactic use of methotrexate associated with an increased or decreased risk of developing aGVHD?
- Platelet recovery analysis:
    - Is recovery of normal platelet levels associated with improved disease-free survival, or a decreased risk of relapse?

This investigation was exploratory, aiming to identify factors that may inform patient prognosis and guide post-transplant monitoring and interventions. This dataset enabled the use of survival analysis techniques that are able to accommodate censored data and time-varying covariates. Our analyses leveraged these methods to give relevant insights into which factors are most strongly associated with specific patient outcomes following bone marrow transplantation.

Methods

While we acknowledge and consider that a p-value of 0.05 is conventionally statistically significant, due to the relatively small size sample (137 enrolled patients), a p-value of ≤ 0.10 was also considered significant for the purposes of this exploratory investigation. All data analysis and table/figure development were conducted using R software.

Baseline and descriptive analyses methods:

A Kaplan-Meier estimator was used to describe disease-free survival (DFS) for all transplanted patients. DFS was defined as the time from transplant to relapse, death, or last follow-up, with patients right censored at last follow-up. Kaplan-Meier curves and associated summary statistics computed, and survival curves were plotted to visualize the decline in disease-free survival over follow-up.

Baseline characteristics were summarized using descriptive statistics stratified by disease group and by FAB classification. Disease group (ALL, AML low risk, AML high risk) and FAB (Other vs FAB 4/5 AML) were treated as categorical variables, and continuous covariates (age, donor age, waiting time, etc.) were compared using one-way tests, while categorical covariates (sex, CMV status, MTX use) were compared using chi-square tests, as implemented in *Table 1*.

A multivariable Cox proportional hazards model was fitted to assess whether baseline characteristics were associated with disease-free survival (DFS). The outcome was DFS time with an event defined as relapse or death. All baseline covariates were entered simultaneously as predictors: age, disease group (ALL, AML low risk, AML high risk), FAB classification (Other vs FAB 4/5 AML), patient sex, patient and donor CMV status, donor age, donor sex, waiting time to transplant, hospital, and prophylactic methotrexate (MTX) use.

aGVHD analyses methods:

We modeled DFS and relapse using time-dependent Cox models, treating aGVHD as a time-varying covariate. DFS was defined as time from transplant to relapse or death, with censoring at last known disease-free status. We fit both unadjusted and adjusted models controlling for age, CMV, FAB, disease group, and MTX.

We restricted to the subset of patients who experienced aGVHD ($n = 26$), and fit Cox proportional hazards model using baseline covariates (patient age, CMV serostatus, donor age, donor CMV status, FAB classification, disease group, MTX use, and wait time between diagnosis and transplant).

We estimated the effect of methotrexate (MTX) use on time to aGVHD using Kaplan-Meier (KM) curves stratified by MTX group, and Cox proportional hazards model adjusted for relevant baseline covariates (age, CMV, FAB, and disease group).

Platelet recovery analysis methods:

Time-dependent Cox proportional hazards models were used to assess whether platelet recovery was associated with DFS and risk of relapse. We created time-dependent datasets, with platelet recovery included as a time-dependent covariate. DFS was defined as the time from transplant to relapse or death, and then relapse was analyzed as a cause-specific event, with deaths without relapse censored. We fit both unadjusted models, and models adjusted for age, CMV, FAB subtype, disease group, and methotrexate prophylaxis, to account for potential confounding effects on DFS and relapse.

To assess the robustness of the association between platelet recovery and relapse, we conducted sensitivity analyses using competing-risk approaches. Fine-Gray subdistribution hazard models accounted for non-relapse mortality as a competing event, and Proportional Odds cumulative incidence function (PO-CIF) models evaluated the effect of platelet recovery on the cumulative probability of relapse over time. These methods complement the Cox model. Additionally, landmark analyses at day 30 (n=133 patients) examined whether early platelet recovery was associated with subsequent relapse while avoiding bias from events occurring before recovery.

We used Fine-Gray hazard models to assess whether platelet recovery was associated with the cumulative incidence of relapse, accounting for death as a competing event. Platelet recovery was evaluated in an ever-recovered model, where recovery at any time post-transplant was coded as “ever recovered,” and in a landmark model, including only patients who survived at least 30 days post-transplant to reduce potential immortal-time bias. Adjusted models included the same covariates as the platelet recovery Cox model. Unadjusted models included only platelet recovery to provide a crude assessment of its association with relapse.

We used PO-CIF to evaluate the association between platelet recovery and relapse after allogeneic transplant, accounting for non-relapse mortality as a competing risk. Platelet recovery was assessed overall and at a day 30 landmark. Models were fit with KM-based censoring. Adjusted models included the same set of covariates used in the previous adjusted Cox and Fine-Gray platelet recovery models; unadjusted models included only platelet recovery.

Results and Discussion

Baseline and descriptive analyses results and discussion:

The Kaplan-Meier estimate (see *Figure 1*) is based on 137 patients, among whom 83 experienced relapse or death during follow-up. The estimated median disease-free survival time is 481 days after transplant, with a fairly wide 95% confidence interval from 381 to 1063 days, reflecting substantial censoring and variability in follow-up. At very early times (e.g., within the first 2-3 months), the DFS probability drops quickly from 1.00 to around 0.85-0.90, indicating that a sizeable fraction of patients relapse or die soon after transplant; thereafter, the curve declines more gradually and stabilizes around 0.35-0.40 at later times, suggesting that roughly one-third of patients remain alive and disease-free several years post-transplant.

Across the three disease groups, age differed significantly: patients with ALL were youngest on average (mean 24.4 years), whereas those with AML low and high risk were older (means 29.4 and 30.4 years; $p = 0.009$). Wait time to transplant also varied markedly, with ALL patients having the longest mean wait (477 days) compared with 138 days in AML low risk and 269 days in AML high risk ($p < 0.001$). Sex and CMV status showed more modest differences: the proportion male ranged from about 53% to 68% across groups ($p = 0.33$), and CMV positivity increased from 39.5% in ALL to 60% in AML high risk ($p = 0.17$). Use of prophylactic methotrexate differed by disease group, with MTX being less common in AML groups (22%-24%) than in ALL (45%; $p = 0.045$), and FAB subtype was strongly associated with disease group: all ALL patients were “Other” FAB, whereas 33% of AML low risk and 60% of AML high risk were FAB 4/5 ($p < 0.001$). Donor age, donor sex, donor CMV, and hospital showed no strong evidence of differences across disease groups (see *Table 1*).

When stratified by FAB (Other vs FAB 4/5 AML), most baseline characteristics were broadly similar between groups. Mean patient age (28.6 vs 27.9 years), donor age (29.0 vs 27.0 years), wait time (309 vs 206 days), sex distribution (61% vs 53% male), and patient CMV status (48% vs 53% CMV positive) did not differ significantly (all $p = 0.12$).

There was some suggestion that donors for FAB 4/5 AML patients were less often CMV positive (31% vs 48%; $p = 0.064$) and that FAB 4/5 AML patients were less likely to receive MTX prophylaxis (18% vs 35%; $p = 0.063$), although these differences did not reach conventional significance. As expected, the FAB variable itself perfectly separated the two strata (by definition). Overall, disease group captured more pronounced differences in baseline covariates (especially age, wait time, MTX use, and FAB subtype) than the FAB classification alone.

Several baseline variables showed statistically significant associations with DFS. Compared with ALL, patients with AML low-risk disease had substantially better DFS (HR 0.40, 95% CI 0.20-0.82, p = 0.013), whereas AML high-risk disease did not differ clearly from ALL (HR 0.87, 95% CI 0.44-1.72, p = 0.69). FAB 4/5 AML was associated with worse DFS than other FAB subtypes (HR 2.48, 95% CI 1.43-4.31, p = 0.0013), indicating more than double the hazard of relapse or death. Patients treated at centers with higher hospital codes had lower hazards (HR per unit increase 0.66, 95% CI 0.50-0.88, p = 0.005), suggesting between-center differences in DFS. MTX prophylaxis was associated with higher hazard of relapse or death (HR 2.10, 95% CI 1.13-3.90, p = 0.018), though this may reflect confounding by indication rather than a harmful effect. Age, sex, CMV status, donor characteristics, and waiting time did not show strong independent associations with DFS in this model (all p > 0.05). Overall, the global tests (likelihood ratio, Wald, and score) were highly significant (p = 0.004), and the model's concordance of about 0.68 indicates moderate ability of baseline variables to discriminate between patients with better versus worse disease-free survival.

aGVHD analyses results and discussion:

As shown in *Table 2*, in the unadjusted time-dependent Cox model, the hazard ratio (HR) for DFS associated with aGVHD was 1.26 (95% CI: 0.73-2.18, p = 0.413), suggesting no significant difference in DFS between patients with and without aGVHD. In the adjusted model, the HR for aGVHD was 1.35 (95% CI: 0.76-2.40, p = 0.309), still indicating no statistically significant association. Among baseline covariates, FAB was significantly associated with lower DFS (HR = 2.26, p = 0.003), while AML Low Risk disease group showed a higher survival (HR = 0.40, p = 0.011) compared to ALL.

For relapse outcome, the unadjusted analysis estimated the HR for aGVHD at 0.67 (95% CI: 0.26-1.70, p = 0.398), showing no significant evidence that aGVHD is associated with a decreased risk of relapse. In the adjusted model, the estimated HR for aGVHD remained non-significant (HR = 0.72, 95% CI: 0.28-1.88, p = 0.502). However, FAB was strongly associated with higher relapse risk (HR = 3.87, p = 0.002), and AML Low Risk again showed a significantly decreased risk of relapse (HR = 0.20, p = 0.003).

Overall, occurrence of aGVHD after transplantation was not significantly associated with improved disease-free survival or with decreased risk of relapse, so it cannot be an important prognostic event. But FAB subtype and disease group showed their prognostic power.

The results summarized in *Table 3* indicate that MTX use was significantly associated with lower DFS (HR = 8.92, 95% CI: 1.42-56.02, p = 0.020), indicating a higher risk of relapse or death among MTX users. Also, wait time between diagnosis and transplant showed a statistically meaningful association (HR = 1.0012, 95% CI: 1.000-1.002, p = 0.060), but with a small effect size. Additionally, donor age (HR = 1.11, p = 0.068) and disease group (AML High Risk vs. ALL: HR = 12.24, p = 0.084) also showed significant associations with lower survival. Note that due to small sample size (n = 26) and number of events (n = 16), these findings require validation in larger cohorts.

Figure 2 shows the unadjusted survival functions for time from transplant until onset of aGVHD by methotrexate use, which suggested a slightly higher survival among MTX users, though the confidence intervals overlapped substantially.

Table 4 summarizes the results of the Cox proportional hazards model for time to onset of aGVHD. HR for MTX was 0.55 (95% CI: 0.21-1.44, p = 0.222), indicating no statistical significance for association with decreased risk of developing aGVHD. Age was significantly associated with increased risk (HR = 1.07, 95% CI: 1.03-1.12, p = 0.0017), suggesting that older patients were more likely to develop aGVHD. Disease group was also significantly associated. Patients with AML High Risk had a significantly lower risk of developing aGVHD (HR = 0.26, 95% CI: 0.08-0.91, p = 0.035), compared to patients with ALL. Overall, there is no statistical evidence that MTX use is associated with decreased risk of aGVHD in this cohort. Age and disease group showed strong influence on aGVHD developing. As a diagnostic check, a proportional hazards (PH) assumption test using Schoenfeld residuals showed that global tests for all do not violate, so PH assumption appeared to hold.

In summary, no evidence was found that aGVHD is associated with improved disease-free survival or decreased relapse risk. Among aGVHD patients, MTX, donor age and high-risk disease group showed the associations with DFS, but the analysis was limited by small sample size. MTX use was not associated with decreased risk of developing aGVHD after adjusting for confounders. Limitations to the aGVHD analyses include small subgroups, resulting in limited power.

Platelet recovery analysis results and discussion:

In the unadjusted time-dependent Cox model analysis, platelet recovery was strongly associated with improved DFS (HR = 0.28, 95% CI: 0.15-0.53, $p < 0.001$). In the adjusted model, platelet recovery remained significantly associated with DFS (HR = 0.36, 95% CI: 0.19-0.69, $p = 0.002$). In addition, FAB subtype (HR = 2.20, 95% CI: 1.27-3.81, $p = 0.005$) and AML Low Risk disease group (HR = 0.41, 95% CI: 0.21-0.82, $p = 0.011$) appeared to be significant independent predictors of DFS. Platelet recovery was not significantly associated with relapse in neither unadjusted (HR = 0.75, 95% CI: 0.22-2.52, $p = 0.64$), nor adjusted analyses (HR = 1.16, 95% CI: 0.34-3.95, $p = 0.82$). FAB subtype (HR = 3.86, 95% CI: 1.67-8.93, $p = 0.002$) and AML Low Risk disease group (HR = 0.20, 95% CI: 0.07-0.60, $p = 0.004$) also appeared to significant independent predictors, of relapse risk (see *Table 5*).

In the time-dependent Cox models, recovery of normal platelet levels after transplant was strongly associated with improved DFS, but did not independently reduce relapse risk. FAB subtype and AML Low Risk disease group were also independently associated with both DFS and relapse. While other factors such as age, CMV serostatus, and MTX were considered, they were not significant in these models. Overall, platelet recovery appears to be an important marker of post-transplant disease-free recovery and survival, although disease characteristics may also play a role in relapse risk.

For the Fine-Gray analysis, in the adjusted ever-recovered model, platelet recovery was not significantly associated with relapse (sHR = 2.21, 95% CI 0.67-7.34, $p = 0.20$). In the adjusted 30-day landmark model, platelet recovery showed a non-significant trend toward lower relapse risk (sHR = 0.77, 95% CI 0.36-1.65, $p = 0.51$). Among covariates, AML Low-Risk disease was associated with a significantly lower relapse incidence, and higher FAB score was associated with higher relapse incidence. The other covariates were not significant. In the unadjusted Fine-Gray models, platelet recovery at any time was not significantly associated with relapse (sHR = 1.97, 95% CI 0.60-6.51, $p = 0.27$). In the unadjusted 30-day landmark analysis, platelet recovery demonstrated a marginally significant trend toward lower relapse risk (sHR = 0.56, 95% CI 0.29-1.10, $p = 0.094$) (see *Table 6*).

In the Fine-Gray models, overall, platelet recovery was not robustly associated with relapse when accounting for competing risks, although early platelet recovery may be tentatively protective. The associations for AML Low-Risk disease and FAB score remained consistent. The landmark analyses, which account for potential time-dependent bias, did suggest that early platelet recovery may be tentatively protective.

In the PO-CIF analyses, most associations between platelet recovery and relapse (platelet recovery overall and in adjusted landmark models) were not statistically significant. The only marginally significant association was observed in the unadjusted landmark model (OR = 0.462, $p = 0.067$), suggesting a lower odds for relapse among patients with platelet recovery by day 30. Other covariates such as low-risk AML and FAB classification were significant, while cmv, age, male, and MTX were consistently not significant (see *Table 7*). Note that there was minor lack-of-fit for AML Low Risk in some goodness-of-fit tests.

The findings from the sensitivity analyses were generally consistent with the primary time-varying Cox models for relapse, which did not demonstrate a significant association. In contrast, platelet recovery was significantly associated with improved disease-free survival in the primary Cox models, highlighting that the effect is more clearly observed for DFS than for relapse specifically. Overall, the results from this analysis support the conclusion that platelet recovery is linked to better DFS, while its effect on relapse remains suggestive but not statistically robust. Limitations include the relatively small number of events for relapse, wide confidence intervals, and potential residual confounding.

Limitations to this investigative study overall included the use of observational data, which may have resulted in possible unmeasured confounding, and a small study population. Future studies may benefit from larger cohorts for increased statistical power, to further investigate the factors associated with patient prognosis following transplant.

Contributions

We divided the work into three sets, and each group member selected a set. This breakdown kept related questions together and allowed each group member a balanced mix of descriptive and analytical work. Key contributions by group members:

- Preetika Banerjee: Set 1: baseline and descriptive analyses (questions 1, 2, and 3)
- Luyi Tong - Set 2: aGVHD analyses (questions 4, 5 and 6)
- Deborah Foster - Set 3: platelet recovery analysis and paper integration (question 7, introduction, integrate findings across sections)

Tables and Figures

Baseline and descriptive analyses tables and figures:

Figure 1: Median Disease-Free Survival Following Transplant

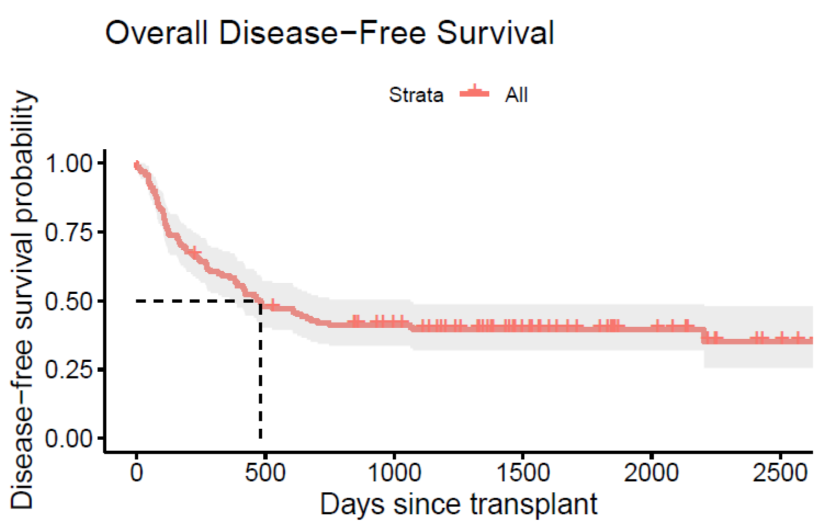

Table 1.a: Descriptive statistics for fab classification, age, male, cmv, donorage, waittime stratified by disease group

Table 1.b: Descriptive statistics for disease group, age, male, cmv, donorage, waittime stratified by fab classifications

Stratified by disgroup

	level	ALL	AML Low Risk	AML High Risk	p	test
n		38	54	45		
age (mean (SD))		24.42 (7.30)	29.41 (8.76)	30.44 (11.22)	0.009	
donorage (mean (SD))		26.79 (8.93)	28.07 (9.24)	29.93 (12.06)	0.367	
waittime (mean (SD))		477.18 (598.86)	138.06 (74.48)	268.87 (210.70)	<0.001	
male (%)	Female	12 (31.6)	24 (44.4)	21 (46.7)	0.329	
	Male	26 (68.4)	30 (55.6)	24 (53.3)		
cmv (%)	CMV-	23 (60.5)	28 (51.9)	18 (40.0)	0.169	
	CMV+	15 (39.5)	26 (48.1)	27 (60.0)		
donormale (mean (SD))		0.68 (0.47)	0.63 (0.49)	0.62 (0.49)	0.819	
donorcmv (mean (SD))		0.45 (0.50)	0.41 (0.50)	0.42 (0.50)	0.931	
hospital (mean (SD))		1.68 (0.84)	2.19 (1.32)	1.80 (1.14)	0.086	
mtx (%)	No MTX	21 (55.3)	42 (77.8)	34 (75.6)	0.045	
	MTX	17 (44.7)	12 (22.2)	11 (24.4)		
fab (%)	Other	38 (100.0)	36 (66.7)	18 (40.0)	<0.001	
	FAB 4/5 AML	0 (0.0)	18 (33.3)	27 (60.0)		

Stratified by fab

	level	Other	FAB 4/5 AML	p	test
n		92	45		
age (mean (SD))		28.60 (9.48)	27.89 (9.81)	0.685	
donorage (mean (SD))		29.00 (9.67)	26.96 (11.13)	0.271	
waittime (mean (SD))		308.72 (426.93)	206.33 (163.85)	0.123	
male (%)	Female	36 (39.1)	21 (46.7)	0.512	
	Male	56 (60.9)	24 (53.3)		
cmv (%)	CMV-	48 (52.2)	21 (46.7)	0.672	
	CMV+	44 (47.8)	24 (53.3)		
donormale (mean (SD))		0.63 (0.49)	0.67 (0.48)	0.680	
donorcmv (mean (SD))		0.48 (0.50)	0.31 (0.47)	0.064	
hospital (mean (SD))		1.93 (1.12)	1.89 (1.25)	0.828	
mtx (%)	No MTX	60 (65.2)	37 (82.2)	0.063	
	MTX	32 (34.8)	8 (17.8)		
fab (%)	Other	92 (100.0)	0 (0.0)	<0.001	
	FAB 4/5 AML	0 (0.0)	45 (100.0)		

aGVHD analyses tables and figures:

Table 2: Time-Dependent Cox Models for Disease-Free Survival (DFS) and Relapse

Outcome	Covariate	HR	95% CI	Coef	p-value
DFS – Unadjusted Model					
DFS	aGVHD	1.26	(0.73, 2.18)	0.229	0.413
DFS – Adjusted Model					
DFS	aGVHD	1.35	(0.76, 2.40)	0.299	0.309
DFS	Age	1.01	(0.98, 1.04)	0.012	0.398
DFS	CMV	0.89	(0.56, 1.42)	-0.118	0.620
DFS	FAB	2.26	(1.32, 3.90)	0.817	**0.003**
DFS	AML Low Risk (vs ALL)	0.40	(0.20, 0.81)	-0.905	**0.011**
DFS	AML High Risk (vs ALL)	0.95	(0.46, 1.97)	-0.049	0.896
DFS	MTX	1.40	(0.85, 2.30)	0.337	0.185
Relapse – Unadjusted Model					
Relapse	aGVHD	0.67	(0.26, 1.70)	-0.403	0.398
Relapse – Adjusted Model					
Relapse	aGVHD	0.72	(0.28, 1.88)	-0.328	0.502
Relapse	Age	1.01	(0.97, 1.05)	0.011	0.574
Relapse	CMV	1.17	(0.60, 2.28)	0.156	0.648
Relapse	FAB	3.87	(1.68, 8.91)	1.352	**0.002**
Relapse	AML Low Risk (vs ALL)	0.20	(0.07, 0.58)	-1.623	**0.003**
Relapse	AML High Risk (vs ALL)	0.67	(0.23, 1.91)	-0.402	0.452
Relapse	MTX	1.22	(0.61, 2.43)	0.198	0.576

Table 3: Cox Model of DSF Among Patients Who Developed aGVHD

Covariate	HR	95% CI	Coef	p-value
Age	0.98	(0.87, 1.10)	-0.0216	0.716
CMV	0.44	(0.10, 1.96)	-0.8145	0.283
Donor Age	1.11	(0.99, 1.25)	0.1069	0.068
Donor CMV	1.86	(0.47, 7.44)	0.6208	0.380
Wait Time	1.0012	(1.000, 1.002)	0.00119	0.060
FAB	1.43	(0.23, 8.89)	0.3543	0.705
AML Low Risk (vs ALL)	4.11	(0.41, 41.61)	1.4130	0.232
AML High Risk (vs ALL)	12.24	(0.72, 209.03)	2.5048	0.084
MTX	8.92	(1.42, 56.02)	2.1882	**0.020**

Figure 2: Unadjusted Time to aGVHD by MTX Use

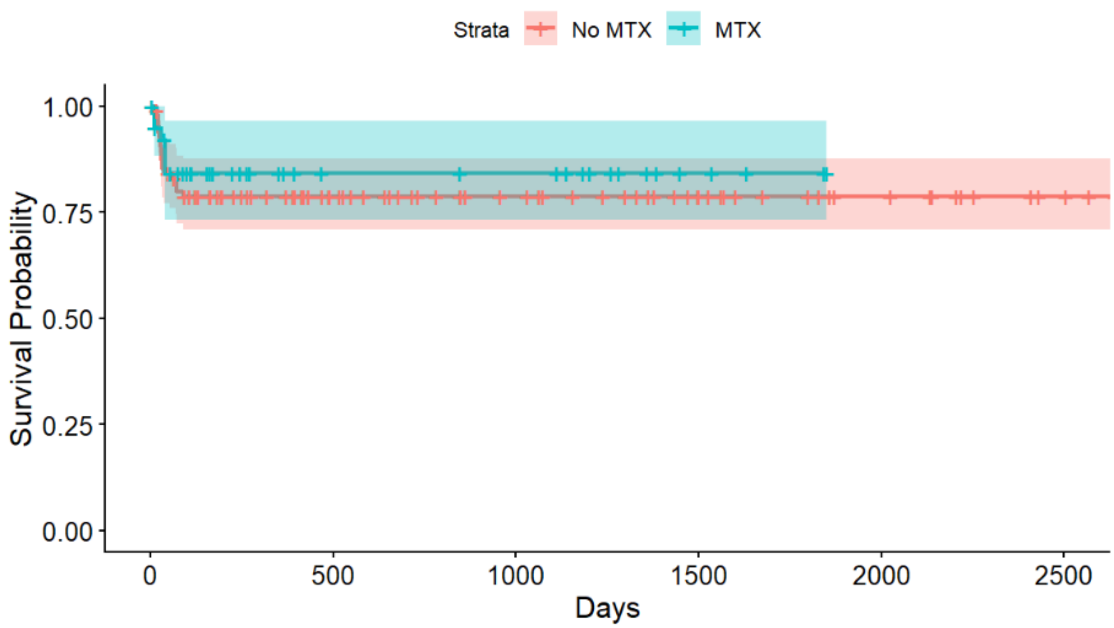

Table 4: Cox Model of Time to aGVHD Onset

Covariate	HR	95% CI	Coef	p-value
MTX	0.55	(0.21, 1.44)	-0.607	0.222
Age	1.07	(1.03, 1.12)	0.070	**0.002**
CMV	0.96	(0.42, 2.20)	-0.041	0.922
FAB	1.33	(0.50, 3.52)	0.286	0.564
AML Low Risk	0.48	(0.17, 1.37)	-0.743	0.168
AML High Risk	0.26	(0.08, 0.91)	-1.339	**0.035**

Platelet recovery analysis tables:

Table 5: Time-Dependent Cox Models of Platelet Recovery, Disease-Free Survival, and Relapse (Unadjusted and Adjusted)

Model	term	HR	95% CI	Coef	p-value
DFS – Unadjusted Model	platelet	0.28	(0.15, 0.53)	-1.276	<0.001
DFS – Adjusted Model	platelet	0.36	(0.19, 0.69)	-1.022	0.002
	age	1.02	(0.99, 1.05)	0.018	0.179
	cmv	0.89	(0.56, 1.42)	-0.114	0.63
	fab	2.20	(1.27, 3.81)	0.790	0.005
	AML Low Risk vs ALL	0.41	(0.21, 0.82)	-0.889	0.011
	AML High Risk vs ALL	0.86	(0.42, 1.73)	-0.156	0.663
	mtx	1.25	(0.75, 2.07)	0.221	0.391
Relapse – Unadjusted Model	platelet	0.75	(0.22, 2.52)	-0.288	0.642
Relapse – Adjusted Model	platelet	1.16	(0.34, 3.95)	0.145	0.817
	age	1.01	(0.97, 1.05)	0.009	0.637
	cmv	1.15	(0.59, 2.24)	0.137	0.688
	fab	3.86	(1.67, 8.93)	1.351	0.002
	AML Low Risk vs ALL	0.20	(0.07, 0.6)	-1.600	0.004
	AML High Risk vs ALL	0.72	(0.25, 2.04)	-0.331	0.534
	mtx	1.26	(0.63, 2.52)	0.232	0.51

Table 6: Fine-Gray Models for Platelet Recovery and Relapse Risk

Model	Covariate	sHR	Lower95	Upper95	p_value
Unadjusted Overall	Platelet Recovery	1.97	0.60	6.51	0.267
Unadjusted Landmark (Day 30)	Platelet Recovery	0.56	0.29	1.10	0.094
Adjusted Overall	platelet_ever	2.21	0.67	7.34	0.195
Adjusted Overall	factor(disgroup)AML Low Risk	0.21	0.07	0.62	0.005
Adjusted Overall	factor(disgroup)AML High Risk	0.61	0.21	1.78	0.369
Adjusted Overall	fab	3.44	1.51	7.87	0.003
Adjusted Overall	age	1.00	0.96	1.04	0.924
Adjusted Overall	male	0.72	0.37	1.38	0.321
Adjusted Overall	cmv	1.55	0.80	3.02	0.198
Adjusted Overall	mtx	1.08	0.53	2.20	0.834
Adjusted Landmark (Day 30)	Platelet Recovery (Landmark Day 30)	0.77	0.36	1.65	0.507
Adjusted Landmark (Day 30)	factor(disgroup)AML Low Risk	0.22	0.08	0.65	0.006
Adjusted Landmark (Day 30)	factor(disgroup)AML High Risk	0.61	0.21	1.81	0.375
Adjusted Landmark (Day 30)	fab	3.32	1.41	7.84	0.006
Adjusted Landmark (Day 30)	age	1.01	0.97	1.05	0.614
Adjusted Landmark (Day 30)	male	0.77	0.40	1.49	0.440
Adjusted Landmark (Day 30)	cmv	1.51	0.80	2.85	0.203
Adjusted Landmark (Day 30)	mtx	0.92	0.44	1.93	0.820

Table 7: Proportional-Odds Cumulative Incidence Function Models for Platelet Recovery and Relapse

Model	Covariate	Estimate	OR	OR 95% CI	p-value
Ever Adjusted	platelet_ever	1.110	3.03	0.70 - 13.12	0.140
	AML Low Risk	-1.780	0.168	0.05 - 0.56	**0.00385**
	AML High Risk	-0.350	0.705	0.21 - 2.33	0.566
	fab	1.560	4.76	1.83 - 12.36	**0.00136**
	cmv	0.381	1.46	0.64 - 3.36	0.369
	age	-0.00279	0.997	0.94 - 1.06	0.923
	male	-0.397	0.672	0.30 - 1.51	0.336
	mtx	0.430	1.54	0.65 - 3.62	0.325
Landmark Adjusted	platelet_at_land	-0.239	0.788	0.31 - 1.97	0.610
	AML Low Risk	-1.760	0.173	0.05 - 0.58	**0.00479**
	AML High Risk	-0.416	0.659	0.19 - 2.31	0.515
	fab	1.502	4.49	1.68 - 11.99	**0.00270**
	cmv	0.329	1.39	0.62 - 3.13	0.427
	age	0.00807	1.008	0.96 - 1.06	0.759
	male	-0.296	0.744	0.33 - 1.68	0.477
Landmark Adjusted	mtx	0.246	1.28	0.53 - 3.08	0.583
Ever Unadjusted	platelet_ever	0.738	2.09	0.53 - 8.19	0.290
Landmark Unadjusted	platelet_at_land	-0.773	0.462	0.20 - 1.06	**0.0673**

Appendix:

Use of AI Tools

For question 1, 2 and 3, AI tools (ChatGPT) were leveraged to help assist in R code development for tables and figures

For question 7, AI tools (ChatGPT) were leveraged to help assist in R code development for tables.

R Code

```
---
title: "Biostat 537 - PROJECT"
subtitle: "Due Date: Friday, December 9th, 2025"
author:
  - "PREETIKA BANERJEE"
date: today
date-format: "12 05, 2025"
format: pdf
---
```

DIRECTIVES
Utilizing the available dataset, address the following scientific questions as appropriately as possible, and describe your findings in a scientific report.

````
```{r}
## Load libraries
library(survival)   # core survival methods
library(survminer)  # nicer KM plots
library(dplyr)      # data manipulation
library(ggplot2)    # plotting

## Load BMT data
bmt <- read.csv("bmt.csv")

## Create basic survival objects used in Q1-3
dfs_surv    <- Surv(bmt$tdfs, bmt$deltadfs)  # disease-free survival
overall_surv <- Surv(bmt$ts,  bmt$deltas)   # overall survival

## Factor labels for disease group
bmt$disgroup <- factor(
  bmt$disgroup,
  levels = c(1, 2, 3),
  labels = c("ALL", "AML Low Risk", "AML High Risk")
)


```
````

1. Provide an estimate of disease-free survival time for patients enrolled in this study. What are the main characteristics of this summary?

```{r}
## Overall disease-free survival (DFS) for all patients

# KM fit for DFS
km_dfs <- survfit(Surv(tdfs, deltadfs) ~ 1, data = bmt)

# Basic summary in the console: n, events, median, CI
print(km_dfs)              # includes median DFS and 95% CI

# More detailed survival summary
summary(km_dfs)

# Plot the DFS curve
ggsurvplot(
  km_dfs,
  conf.int = TRUE,
  xlab = "Days since transplant",
  ylab = "Disease-free survival probability",
  title = "Overall Disease-Free Survival",
  surv.median.line = "hv"     # add horizontal/vertical lines at median
)

```

A Kaplan-Meier estimator was used to describe disease-free survival (DFS) for all transplanted patients.We defined a right-censored survival object Surv(tdfs, deltadfs), where tdfs is the time from transplant to relapse, death, or last follow-up, and deltadfs = 1 indicates relapse or deat. This object was passed to `survfit(Surv(tdfs, deltadfs) ~ 1, data = bmt)` to obtain the nonparametric DFS curve and associated summary statistics, and the curve was plotted to visualize the decline in disease-free survival over follow-up.

The Kaplan-Meier estimate is based on 137 patients, among whom 83 experienced relapse or death during follow-up. The estimated median disease-free survival time is 481 days after transplant, with a fairly wide 95% confidence interval from 381 to 1063 days, reflecting substantial censoring and variability in follow-up. At very early times (e.g., within the first 2-3 months), the DFS probability drops quickly from 1.00 to around 0.85-0.90, indicating that a sizeable fraction of patients relapse or die soon after transplant; thereafter, the curve declines more gradually and stabilizes around 0.35-0.40 at later times, suggesting that roughly one-third of patients remain alive and disease-free several years post-transplant.

2. How do patients in different disease groups or in different FAB classifications compare to each other with respect to other available baseline measurements?

```{r}

##  bmt$disgroup; it is already a factor:
str(bmt$disgroup)
table(bmt$disgroup, useNA = "ifany")

## Only (re)code the other factors
bmt$fab  <- factor(bmt$fab,  levels = c(0, 1), labels = c("Other", "FAB 4/5 AML"))
bmt$mtx  <- factor(bmt$mtx,  levels = c(0, 1), labels = c("No MTX", "MTX"))
```

```
bmt$cmv  <- factor(bmt$cmv,  levels = c(0, 1), labels = c("CMV−", "CMV+"))
bmt$male <- factor(bmt$male, levels = c(0, 1), labels = c("Female", "Male"))

library(tableone)

vars <- c("age", "donorage", "waittime", "male", "cmv", "donormale",
        "donorcmv", "hospital", "mtx", "fab")

tab_disgroup <- CreateTableOne(vars = vars, strata = "disgroup", data = bmt)
print(tab_disgroup, showAllLevels = TRUE, quote = FALSE, noSpaces = TRUE)

tab_fab <- CreateTableOne(vars = vars, strata = "fab", data = bmt)
print(tab_fab, showAllLevels = TRUE, quote = FALSE, noSpaces = TRUE)

```

Baseline characteristics were summarized using descriptive statistics stratified by disease group and by FAB classification. Disease group (ALL, AML low risk, AML high risk) and FAB (Other vs FAB 4/5 AML) were treated as categorical variables, and continuous covariates (age, donor age, waiting time, etc.) were compared using one-way tests, while categorical covariates (sex, CMV status, MTX use) were compared using chi-square tests, as implemented in `CreateTableOne`.

By disease group

Across the three disease groups, age differed significantly: patients with ALL were youngest on average (mean 24.4 years), whereas those with AML low and high risk were older (means 29.4 and 30.4 years; $p = 0.009$). Wait time to transplant also varied markedly, with ALL patients having the longest mean wait (477 days) compared with 138 days in AML low risk and 269 days in AML high risk ($p < 0.001$).

Sex and CMV status showed more modest differences: the proportion male ranged from about 53% to 68% across groups ($p = 0.33$), and CMV positivity increased from 39.5% in ALL to 60% in AML high risk ($p = 0.17$). Use of prophylactic methotrexate differed by disease group, with MTX being less common in AML groups (22%-24%) than in ALL (45%; $p = 0.045$), and FAB subtype was strongly associated with disease group: all ALL patients were “Other” FAB, whereas 33% of AML low risk and 60% of AML high risk were FAB 4/5 ($p < 0.001$). Donor age, donor sex, donor CMV, and hospital showed no strong evidence of differences across disease groups.

By FAB classification

When stratified by FAB (Other vs FAB 4/5 AML), most baseline characteristics were broadly similar between groups. Mean patient age (28.6 vs 27.9 years), donor age (29.0 vs 27.0 years), wait time (309 vs 206 days), sex distribution (61% vs 53% male), and patient CMV status (48% vs 53% CMV positive) did not differ significantly (all $p \geq 0.12$).

There was some suggestion that donors for FAB 4/5 AML patients were less often CMV positive (31% vs 48%; $p = 0.064$) and that FAB 4/5 AML patients were less likely to receive MTX prophylaxis (18% vs 35%; $p = 0.063$), although these differences did not reach conventional significance. As expected, the FAB variable itself perfectly separated the two strata (by definition). Overall, disease group captured more pronounced differences in baseline covariates (especially age, wait time, MTX use, and FAB subtype) than the FAB classification alone.

3. Are any of the measured baseline variables associated with differences in disease-free survival?
```{r}

# Disease-free survival outcome
dfs_surv <- Surv(bmt$tdfs, bmt$deltadfs)

# Fit Cox model with baseline covariates
cox_dfs_full <- coxph(
  dfs_surv ~ age + disgroup + fab + male + cmv +
    donorage + donormale + donorcmv +
    waittime + hospital + mtx,
  data = bmt
)

summary(cox_dfs_full)
```
A multivariable Cox proportional hazards model was fitted to assess whether baseline characteristics were associated with disease-free survival (DFS). The outcome was DFS time (`tdfs`) with an event defined as relapse or death (`deltadfs = 1`), encoded as `Surv(tdfs, deltadfs)`. All baseline covariates were entered simultaneously as predictors: age, disease group (ALL, AML low risk, AML high risk), FAB classification (Other vs FAB 4/5 AML), patient sex, patient and donor CMV status, donor age, donor sex, waiting time to transplant, hospital, and prophylactic methotrexate (MTX) use.

Several baseline variables showed statistically significant associations with DFS. Compared with ALL, patients with AML low-risk disease had substantially better DFS (HR 0.40, 95% CI 0.20-0.82, p = 0.013), whereas AML high-risk disease did not differ clearly from ALL (HR 0.87, 95% CI 0.44-1.72, p = 0.69). FAB 4/5 AML was associated with worse DFS than other FAB subtypes (HR 2.48, 95% CI 1.43-4.31, p = 0.0013), indicating more than double the hazard of relapse or death. Patients treated at centers with higher hospital codes had lower hazards (HR per unit increase 0.66, 95% CI 0.50-0.88, p = 0.005), suggesting between-center differences in DFS. MTX prophylaxis was associated with higher hazard of relapse or death (HR 2.10, 95% CI 1.13-3.90, p = 0.018), though this may reflect confounding by indication rather than a harmful effect. Age, sex, CMV status, donor characteristics, and waiting time did not show strong independent associations with DFS in this model (all $p > 0.05$). Overall, the global tests (likelihood ratio, Wald, and score) were highly significant ($p \leq 0.004$), and the model's concordance of about 0.68 indicates moderate ability of baseline variables to discriminate between patients with better versus worse disease-free survival.

---
title: "537BMTQ4-6"
author: "Luyi Tong"
date: "2025-12-04"
output: html_document

```{r}
library(survival)
library(survminer)
library(dplyr)
library(ggplot2)
```

````
```{r}
bmt <- read.csv("bmt.csv")
```

# baseline survival
```{r}
dfs_surv <- Surv(bmt$tdfs, bmt$deltadfs)      # disease-free survival
relapse_surv <- Surv(bmt$tdfs, bmt$deltar)    # relapse survival
agvhd_surv <- Surv(bmt$ta, bmt$deltaa)        # time to aGVHD
bmt$disgroup <- factor(bmt$disgroup,
                   levels = c(1, 2, 3),
                   labels = c("ALL", "AML Low Risk", "AML High Risk"))
```


# Q4: is aGVHD associated with improved DFS and lower relapse?

## DFS (unadjusted)
```{r}
bmt_td_dfs <- tmerge(data1 = bmt, data2 = bmt, id = id,
              tstart = 0, tstop = tdfs,
              event = event(tdfs, deltadfs),
              agvhd = tdc(ta))

# time-dependent cox model
cox_td_dfs <- coxph(Surv(tstart, tstop, event) ~ agvhd, data = bmt_td_dfs)
summary(cox_td_dfs)
```

## DFS (adjusted)
```{r}
cox_td_dfs_adj <- coxph(Surv(tstart, tstop, event) ~ agvhd + age + cmv + fab + disgroup + mtx, data =
bmt_td_dfs)
summary(cox_td_dfs_adj)

```


## relapse (unadjusted)
```{r}
bmt_td_relapse <- tmerge(data1 = bmt, data2 = bmt, id = id,
                  tstart = 0, tstop = tdfs,
                  relapse = event(tdfs, deltar),
                  agvhd = tdc(ta))

# time-dependent cox model
cox_td_relapse <- coxph(Surv(tstart, tstop, relapse) ~ agvhd, data = bmt_td_relapse)
summary(cox_td_relapse)
```
````

````
## relapse (adjusted)
```{r}
cox_td_relapse_adj <- coxph(Surv(tstart, tstop, relapse) ~ agvhd + age + cmv + fab + disgroup + mtx,
data = bmt_td_relapse)
summary(cox_td_relapse_adj)

```




# Q5: Among patients with aGVHD, what baseline factors predict DFS?
```{r}
# patients with aGVHD
bmt_agvhd <- bmt %>% filter(deltaa == 1)

dfs_surv_agvhd <- Surv(bmt_agvhd$tdfs, bmt_agvhd$deltadfs)
cox_agvhd <- coxph(dfs_surv_agvhd ~ age + cmv + donorage + donorcmv +
                                        waittime + fab + disgroup + mtx,
                        data = bmt_agvhd)
summary(cox_agvhd)
```


# Q6: MTX ~ aGVHD?

## KM plot
```{r}
km_agvhd_mtx <- survfit(agvhd_surv ~ mtx, data = bmt)
ggsurvplot(km_agvhd_mtx, conf.int = TRUE,
        title = "Unadjusted Time to aGVHD by MTX Use",
        legend.labs = c("No MTX", "MTX"),
        xlab = "Days", ylab = "Survival Probability")
```

## adjusted Cox model
```{r}
cox_agvhd_mtx <- coxph(Surv(ta, deltaa) ~ mtx + age + cmv + fab + disgroup, data = bmt)
summary(cox_agvhd_mtx)

```





# PH assumption check
```{r}
cox.zph(cox_agvhd_mtx)
```

# Deborah Foster, 12/6/2025
# Final Project BIOST 537 - Question 7
````

```
# Platelet Recovery and Disease-Free Survival / Relapse
# R code below includes:
# primary Cox, cause-specific Cox, Fine-Gray, proportional-odds CIF, tables/figures

# Read in dataset, bmt
# bmt <- read.csv("bmt.csv")

# Question 7: Based on the available data, is recovery of normal platelet levels associated with improved
disease-
# free survival? Is it associated with a decreased risk of relapse?

library(dplyr)
library(survival)
library(cmprsk)
library(timereg)
library(mstate)

# -----------------------------
#  Prep dataset
# -----------------------------
bmt <- bmt %>%
  mutate(
    id     = as.integer(id),
    tdfs   = as.numeric(tdfs),
    ts     = as.numeric(ts),
    tp     = as.numeric(tp),
    ta     = as.numeric(ta),
    deltadfs = as.numeric(deltadfs),
    deltar   = as.numeric(deltar),
    deltas   = as.numeric(deltas)
  ) %>%
  mutate(
    platelet_ever = ifelse(!is.na(tp) & deltap==1, 1, 0),
    agvhd_ever   = ifelse(!is.na(ta) & deltaa==1, 1, 0)
  )
# Note: opted to exclude agvhd as a confounder in sensitivity analysis since it is not shown to
independently predict DFS or relapse.
####################################
# landmark determination for sensitivity:
# Examine time to platelet recovery
summary(bmt$tp)           # basic stats: min, median, max
   # Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   # 0.00   14.00   18.00   49.07   27.00 1298.00
hist(bmt$tp, main="Time to platelet recovery", xlab="Days", breaks=20)
# visual check

# Candidate landmark times (in 10-day increments)
candidate_days <- seq(10, 100, by=10)

# Compute number of patients at risk (alive and disease-free) at each candidate day
at_risk <- sapply(candidate_days, function(day) {
```

```
  sum(bmt$tdfs > day)
})

# Compute number of patients who have recovered platelets by that day
platelet_recovered <- sapply(candidate_days, function(day) {
  sum(!is.na(bmt$tp) & bmt$deltap == 1 & bmt$tp <= day)
})

# Combine into a table
landmark_table <- data.frame(
  LandmarkDay = candidate_days,
  PatientsAtRisk = at_risk,
  PlateletRecovered = platelet_recovered
)

print(landmark_table)
#    LandmarkDay PatientsAtRisk PlateletRecovered
# 1          10            134                 7
# 2          20            133                76
# 3          30            133               102
# 4          40            131               109
# 5          50            127               111
# 6          60            125               116
# 7          70            123               119
# 8          80            117               119
# 9          90            115               119
# 10        100            113               120

# review: plot to visualize
library(ggplot2)
ggplot(landmark_table, aes(x=LandmarkDay)) +
  geom_line(aes(y=PatientsAtRisk, color="At Risk")) +
  geom_line(aes(y=PlateletRecovered, color="Recovered")) +
  labs(y="Number of Patients", color="Legend",
       title="Candidate Landmark Times for Platelet Recovery") +
  theme_minimal()
# Observation: at about day 30, most recoveries have occurred, the number of recoveries increases slowly,
# and there are plenty of patients at risk to maintain statistical power
############################

# baseline survival check
dfs_surv <- Surv(bmt$tdfs, bmt$deltadfs)      # disease-free survival
relapse_surv <- Surv(bmt$tdfs, bmt$deltar)    # relapse survival
platelet_surv <- Surv(bmt$tp, bmt$deltap)       # time to platelet recovery

summary(dfs_surv)
summary(relapse_surv)
summary(platelet_surv)

######################
```

```
# ## Per patient unadjusted Cox model, platelet recovery - ignoring TVCs
# # Releapse
# cox_relapse_unadj_patient <- coxph(Surv(tdfs, deltar) ~ platelet_ever, data = bmt)
#
# summary(cox_relapse_unadj_patient)
#
# # Unadjusted DFS model (per patient, ignoring TVCs)
# cox_dfs_unadj_patient <- coxph(
#   Surv(tdfs, deltadfs) ~ platelet_ever,  # platelet_ever = ever recovered normal platelets
#   data = bmt
# )
#
# summary(cox_dfs_unadj_patient)
################################################################################
# Is recovery of normal platelet levels associated with improved DFS and lower relapse?
# 1. DFS (unadjusted time-dependent Cox model)
#This model evaluates whether platelet recovery—treated as a time-dependent covariate—
#is associated with disease-free survival without adjusting for other patient or transplant characteristics.
bmt$disgroup <- factor(bmt$disgroup, levels = c(1,2,3),
                       labels = c("ALL", "AML Low Risk", "AML High Risk"))

bmt_td_dfs <- tmerge(
  data1 = bmt, data2 = bmt, id = id,
  tstart = 0, tstop = tdfs,
  event = event(tdfs, deltadfs),
  platelet = tdc(tp)   # time at which platelet recovery occurs
)

# unadjusted time-dependent Cox model for DFS
cox_td_dfs <- coxph(Surv(tstart, tstop, event) ~ platelet, data = bmt_td_dfs)
summary(cox_td_dfs)

#2. DFS (adjusted time-dependent Cox model)
#This model assesses whether platelet recovery is associated with improved DFS after
#adjusting for age, CMV status, FAB subtype, underlying disease group, and methotrexate prophylaxis.

cox_td_dfs_adj <- coxph(
  Surv(tstart, tstop, event) ~ platelet + age + cmv + fab + factor(disgroup) + mtx,
  data = bmt_td_dfs
)
summary(cox_td_dfs_adj)

# 3. Relapse (unadjusted time-dependent Cox model)
# This model determines whether platelet recovery is associated with a reduced cause-specific hazard of
relapse,
# without adjusting for covariates.
# Death without relapse is censored.

bmt_td_relapse <- tmerge(
  data1 = bmt, data2 = bmt, id = id,
  tstart = 0, tstop = tdfs,
```

```
  relapse = event(tdfs, deltar),
  platelet = tdc(tp)
)

# unadjusted time-dependent Cox model for relapse
cox_td_relapse <- coxph(Surv(tstart, tstop, relapse) ~ platelet, data = bmt_td_relapse)
summary(cox_td_relapse)

#4. Relapse (adjusted time-dependent Cox model)

#This model evaluates whether platelet recovery is associated with a lower risk of
#relapse after adjustment for key clinical and demographic covariates.

cox_td_relapse_adj <- coxph(
  Surv(tstart, tstop, relapse) ~ platelet + age + cmv + fab + factor(disgroup) + mtx,
  data = bmt_td_relapse
)
summary(cox_td_relapse_adj)

## Cox Tables:
library(dplyr)
library(broom)
library(kableExtra)

# Map disgroup numeric codes to descriptive labels
disgroup_labels <- c(
  "1" = "ALL",
  "2" = "AML Low Risk vs ALL",
  "3" = "AML High Risk vs ALL"
)

# List of models
models <- list(
  "DFS - Unadjusted Model" = cox_td_dfs,
  "DFS - Adjusted Model"   = cox_td_dfs_adj,
  "Relapse - Unadjusted Model" = cox_td_relapse,
  "Relapse - Adjusted Model"   = cox_td_relapse_adj
)

tidy_cox_nice <- function(model, model_name) {
  tidy(model, conf.int = TRUE) %>%
    mutate(
      Outcome = ifelse(grepl("DFS", model_name), "DFS", "Relapse"),
      Model   = model_name,
      # Clean up factor names for disgroup
      term = case_when(
        grepl("factor\\(disgroup\\)AML Low Risk", term)  ~ "AML Low Risk vs ALL",
        grepl("factor\\(disgroup\\)AML High Risk", term) ~ "AML High Risk vs ALL",
        TRUE ~ term
      ),
      HR = round(exp(estimate), 2),
```

```
    `95% CI` = paste0("(", round(exp(conf.low),2), ", ", round(exp(conf.high),2), ")"),
    Coef = round(estimate, 3),
    `p-value` = as.character(ifelse(p.value < 0.001, "<0.001", round(p.value,3)))
  ) %>%
  select(Outcome, Model, term, HR, `95% CI`, Coef, `p-value`)
}


cox_table <- bind_rows(lapply(names(models), function(x) tidy_cox_nice(models[[x]], x)))

cox_table %>%
 kable(format = "html", caption = "Time-Dependent Cox Models for DFS and Relapse") %>%
 kable_styling(full_width = FALSE, position = "center")

# Suppress repeated Outcome and Model values
cox_table_clean <- cox_table %>%
 group_by(Outcome, Model) %>%
 mutate(
  Outcome = ifelse(row_number() == 1, Outcome, ""),
  Model   = ifelse(row_number() == 1, Model, "")
 ) %>%
 ungroup()

# Display nicely
cox_table_clean %>%
 kable(format = "html", caption = "Time-Dependent Cox Models for DFS and Relapse") %>%
 kable_styling(full_width = FALSE, position = "center")

## end of tables
##############################################################################


# -----------------------------
# Fine-Gray subdistribution hazard for relapse
# -----------------------------
bmt_fg <- bmt %>%
 mutate(fstatus = ifelse(deltar==1, 1, ifelse(deltas==1, 2, 0)))

cov_fg <- model.matrix(~ platelet_ever  + factor(disgroup) + fab + age + male + cmv + mtx,
data=bmt_fg)[,-1]

fg_model <- crr(ftime=bmt_fg$tdfs, fstatus=bmt_fg$fstatus, cov1=cov_fg)
summary(fg_model)

# Landmark Fine-Gray (e.g., day 30)
landmark <- 30
bmt_land <- bmt %>%
 filter(tdfs > landmark) %>%
 mutate(
  platelet_at_land = ifelse(!is.na(tp) & deltap==1 & tp <= landmark,1,0),
#   agvhd_at_land   = ifelse(!is.na(ta) & deltaa==1 & ta <= landmark,1,0),#excluding this.
```

```
    time_from_land  = tdfs - landmark,
    fstatus = ifelse(deltar==1, 1, ifelse(deltas==1, 2, 0))
  )

cov_land <- model.matrix(~ platelet_at_land + factor(disgroup) + fab + age + male + cmv + mtx,
data=bmt_land)[,-1]
fg_land <- crr(ftime=bmt_land$time_from_land, fstatus=bmt_land$fstatus, cov1=cov_land)
summary(fg_land)

# -----------------------------
# Unadjusted Fine-Gray subdistribution hazard for relapse
# -----------------------------

# Overall unadjusted model
fg_model_unadj <- crr(
  ftime = bmt_fg$tdfs,
  fstatus = bmt_fg$fstatus,
  cov1 = model.matrix(~ platelet_ever, data=bmt_fg)[,-1]
)
summary(fg_model_unadj)

# Landmark unadjusted model (e.g., day 30)
fg_land_unadj <- crr(
  ftime = bmt_land$time_from_land,
  fstatus = bmt_land$fstatus,
  cov1 = model.matrix(~ platelet_at_land, data=bmt_land)[,-1]
)
summary(fg_land_unadj)

# Table for 4 Fine-Gray models

library(dplyr)
library(kableExtra)

tidy_fg <- function(fg_model, model_name) {
  cov_names <- if(!is.null(rownames(fg_model$coef))) {
    rownames(fg_model$coef)
  } else {
    names(fg_model$coef)
  }

  # Clean up ugly names
  cov_names <- gsub("model.matrix\\(.*\\)\\[, -1\\]1", "Platelet Recovery", cov_names)
  cov_names <- gsub("platelet_at_land", "Platelet Recovery (Landmark Day 30)", cov_names)

  data.frame(
    Model = model_name,
    Covariate = cov_names,
    sHR = exp(fg_model$coef),
    Lower95 = exp(fg_model$coef - 1.96 * sqrt(diag(fg_model$var))),
    Upper95 = exp(fg_model$coef + 1.96 * sqrt(diag(fg_model$var))),
```

```
    p_value = 2 * (1 - pnorm(abs(fg_model$coef / sqrt(diag(fg_model$var)))))
  )
}

# Combine and round
fg_table <- bind_rows(
  tidy_fg(fg_model_unadj, "Unadjusted Overall"),
  tidy_fg(fg_land_unadj, "Unadjusted Landmark (Day 30)"),
  tidy_fg(fg_model, "Adjusted Overall"),
  tidy_fg(fg_land, "Adjusted Landmark (Day 30)")
) %>%
  mutate(
    sHR = round(sHR, 2),
    Lower95 = round(Lower95, 2),
    Upper95 = round(Upper95, 2),
    p_value = round(p_value, 3)
  )

# Render table
fg_table %>%
  kable(format = "html", caption = "Fine-Gray subdistribution hazard models for relapse") %>%
  kable_styling(full_width = F, position = "center")



# -----------------------------
# Proportional-odds CIF
# -----------------------------
po_fit_ever <- tryCatch({
  prop.odds.subdist(Event(tdfs, fstatus) ~ platelet_ever  + factor(disgroup) + fab + cmv + age + male +
mtx,
                    data=bmt_fg, cause=1, cens.model="KM", n.sim=200)
}, error=function(e) { message("PO-CIF failed: ", e$message); NULL })

po_fit_land <- tryCatch({
  prop.odds.subdist(Event(time_from_land, fstatus) ~ platelet_at_land  + factor(disgroup) + fab + cmv +
age + male + mtx,
                    data=bmt_land, cause=1, cens.model="KM", n.sim=200)
}, error=function(e) { message("PO-CIF (landmark) failed: ", e$message); NULL })

po_fit_ever
po_fit_land

# Robustness check
po_fit_ever_unadj <- tryCatch({
  prop.odds.subdist(Event(tdfs, fstatus) ~ platelet_ever,
                    data=bmt_fg, cause=1, cens.model="KM", n.sim=200)
}, error=function(e) { message("PO-CIF failed: ", e$message); NULL })
po_fit_ever_unadj

# Landmark unadjusted Proportional Odds CIF (e.g., day 30)
```

```
po_fit_land_unadj <- tryCatch({
  prop.odds.subdist(Event(time_from_land, fstatus) ~ platelet_at_land,
                    data = bmt_land, cause = 1, cens.model = "KM", n.sim = 200)
}, error = function(e) { message("PO-CIF (landmark unadjusted) failed: ", e$message); NULL })

# View results
po_fit_land_unadj
```